



Cymbal Agentic Suite — System Architecture

Autonomous Customer Lifecycle, Protocol Commerce & Revenue Recovery Engine

● LAYER 1: MULTIMODAL EVIDENCE INGESTION & GROUNDING SOURCES



Places Insights API

Quantitative regional benchmarks, competitor foot-traffic, and rating delta anomalies across retail depots.



Maps Grounding

Qualitative sentiment extraction from customer feedback and geographical proximity fitment resolution.



BigQuery

Regional anomaly detection and customer feedback dossiers on detractors.

● LAYER 2: AUTONOMOUS INTELLIGENCE CORE & DETERMINISTIC GUARDRAIL PIPELINE



Long Horizon Orchestrator (Google ADK 2.5 + Gemini 3.7 Flash)

1

Tier 1: Static Prefix

Role & System Guardrails



2

Tier 2: Context Callback

Store Depots & OKF SOPs

Experience Agent

Gemini 3.7 Flash

- Ingests `customer.survey.submitted`
- Synthesizes Maps Grounding & Ratings
- Dispatches direct review nudges
- Triggers Google Chat HITL card on ≤ 6 NPS

Recovery Agent

Gemini 3.7 Flash

- Monitors `checkout.stalled` (15m delay)
- Calculates bounded recovery discount
- Issues SD-JWT CartMandate (2h TTL)
- Dispatches `commerce.recovery.offer`

Investigator Agent

- Investigates detractor feedback
- Monitors regional sentiment
- Validates feedback authenticity
- Escalates critical issues

● LAYER 3: AUTONOMOUS SURFACES, PROTOCOL ECOSYSTEM & MULTIMODAL MODELS



Cymbal Storefront

Imagen 3 + Veo 2 + Next.js 15

- Imagen 3 Photorealistic Wheel Renders
- Veo 2 360° Fitment Clearance Motion
- Gemini 3.7 Flash Conversational Fitment



Google Chat HITL Cards

Cards v2 Declarative Schema

- Real-Time Detractor Escalation Alerts
- In-Place Action Webhook (Investigate / Dismiss)
- Regional Store Manager Workflows

● CLOUD RUN LIVE DEPLOYED • Google #AllThingsAgenticHackathon Submission • Models: Gemini 3.7 Flash, Gemini 3.7 Thinking, Gemma, Imagen 3, Veo 2, Lyria